INDIAN INSTITUTE OF TECHNOLOGY, GANDHINAGAR
ES 116: Sem II, 2025-2026
Principles and Applications of Electrical Engineering

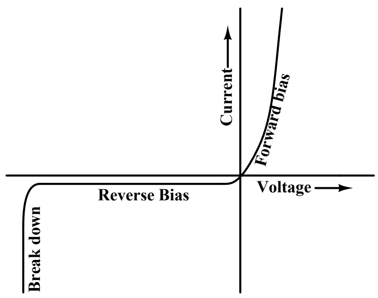
Tutorial-5
(Diode circuits and Applications)

A brief review

Diodes are two terminal devices whose conduction properties are asymmetric with respect to the applied voltage.
Their current-voltage characteristics can be described by the relation

$$I = I_0 \left(e^{qV/k_BT} - 1\right)$$

where V is the voltage across the diode and is positive when the "anode" terminal is made positive with respect to the "cathode" terminal.
(q is the magnitude of electron charge, k_B is the Boltzmann constant and T is the temperature in degrees Kelvin.)



The operating region where V is positive is termed as "forward" bias, while the region where V is negative is called "reverse" bias.

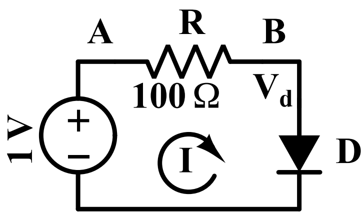
At normal room temperature, k_BT/q is quite small (– about 26 mV). The exponent qV/k_BT is about 30 for an applied bias of 0.8 V. So in foraward bias, the exponential term e^{qV/k_BT} is $\approx 10^{13}$. Thus the exponential is the dominant term and 1 can be neglected.
On the other hand, for a reverse bias of the same magnitude, the exponential term is $\approx 10^{-13}$, which is negligible compared to 1. Therefore the current becomes constant for applied reverse bias when its magnitude exceeds about 100 mV (where the exponent term is $\approx .02$). However, if the magnitude of the reverse bias becomes more than a specific voltage depending on the construction of the diode (called the breakdown voltage), the current increases rapidly with very little change in the reverse bias.

Circuit Analysis with Diodes

The current voltage characteristics of the diode are non-linear and asymmetric. This makes analysis of circuits with diodes somewhat tedious. To illustrate this consider the simple circuit shown below:
Voltage at A = 1.0 V. Let the voltage at B be V_d and the current through the mesh be I. By the diode equation,

$$I = I_0 \left(e^{qV_d/k_BT} - 1\right)$$

Let the value of $I_0 = 10^{-14}$ A. (This is a typical value for silicon diodes).



From the diode equation:

$$V_d = \frac{k_B T}{q} \ln\left(1 + \frac{I}{I_0}\right)$$

Applying KVL, we get

$$1 - 100I - \frac{k_B T}{q} \ln\left(1 + \frac{I}{I_0}\right) = 0$$

This is a transcendental equation in I and cannot be solved in closed form. We have to get the result by assuming an initial guess for I and iterate to get the final solution. Writing

$$I = \frac{1}{100}\left(1 - \frac{k_B T}{q} \ln\left(1 + \frac{I}{I_0}\right)\right)$$

and iterating with a starting guess of $I = 4$ mA, we get successive values of I as:
$3.054, 3.124, 3.118, 3.119, 3.119\,\mathrm{mA}$.
So $I = 3.119$ mA and correspondingly, $V_d = 1 - 100I = 0.6881$ V.

This shows that even a simple circuit is hard to analyse with the accurate current equation of the diode. Therefore, we often use approximate models to represent the diode during circuit analysis.

Ideal Diode model

Here we model the diode as a switch. The switch is “on” (has zero resistance) in the forward region and is “off” (has infinite resistance) in the reverse region. Since the drop across the diode will be zero according to this model, it is useful when the voltages in the circuit are high and the actual drop across the diode can be considered negligible.
In the example considered above, this model will predict a current of 10 mA through the diode – which is quite inaccurate. On the other hand, if the source voltage is 10 V, the accurate solution will give 92.24 mA as the current, while the ideal diode model will predict 100 mA, which is close.

Constant Voltage Drop in Forward Region

The ideal diode model makes circuit analysis quite simple, but it may be considered a bit extreme when voltages in the circuit are relatively low. We notice that the drop across the diode changes very slowly with current. For example, in the diode considered in the example above, the drop across the diode is 0.66 V when the current is 1 mA, it is 0.72 V when the current is 10 mA and 0.78 V when the current is 100 mA. Thus the voltage drop across the diode changes by millivolts even as the current varies over several orders of magnitude.

So in this approximation, we treat the diode as a switch, but with a constant voltage drop in the forward conduction region. For widely used silicon p-n junction diodes, this drop is taken to be around 0.55 to 0.7 V in low current circuits and about 0.8 to 1 V in high current circuits. Germanium diodes have a drop of about 0.2 V, while light emitting diodes (LEDs) have a much higher drop of 1.5 to 2.5 V. Schottky diodes made with metal junctions to semiconductors exhibit much lower drops of about 0.15 to 0.2 V. This drop is often designated as V_d and its value is decided depending on the type of diode.

In the example considered above for a silicon diode, if we take the drop to be 0.65 V, the current will be 3.5 mA, which is close to the accurate solution. For a source voltage of 10 V, it will predict a current of (10 − 0.65)/100 A, which is 93.5 mA, quite close to the accurate solution which is 92.24 mA.

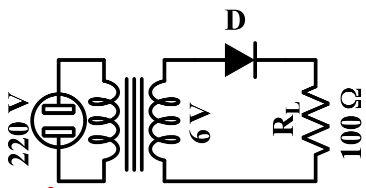
Diode applications

Diodes permit appreciable current flow in only the forward direction, and when the reverse voltage exceeds their breakdown voltage. Some of the applications of diodes are:

- conversion of AC sources to DC – known as rectification,
- to generate predictable output voltages from variable inputs by operating them in the reverse break down region,
- to restrict voltage variation beyond specific voltages – known as clamping or clipping,
- to perform logical functions like OR etc.,
- to emit light – since some diodes made with compound semiconductors emit light when operated in the forward region, and
- to act as variable capacitors whose value can be adjusted by adjusting the reverse voltage across them (known as varactors).

In this tutorial, we'll design and analyse circuits associated with such applications.

Q–1 Consider a diode being used as a rectifier.
A transformer reduces the 220 V 50 Hz mains voltage to 6 V. A diode connects the top of the secondary coil to the load resistor of 100 Ω. We can represent the transformer along with the resistance of the windings by an AC source of 6 V rms with a series resistor of 2 Ω.



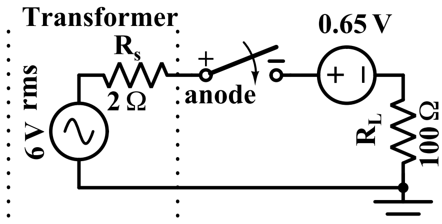
(i.e. $V_D = 0$ across the diode in FB)

a) Using the ideal diode model, draw the equivalent circuit of the rectifier circuit above in the positive and negative half cycles. Draw the waveform of the voltage across the load resistor.

b) What is the peak voltage across the 100 Ω resistor? Compute the average voltage across the resistor over one cycle of AC.

c) We now use the constant forward drop model for the diode.

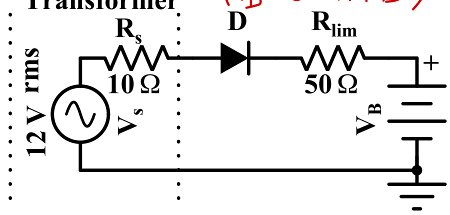
Assume that the diode is equivalent to a switch with a series connected source of 0.65 V which subtracts from the input in the positive half cycle as shown in the figure. The switch is "on" only when the voltage across *it (the switch)* is positive.



Does the diode conduct during the entire positive half-cycle of AC?
What is the peak voltage across the load resistor?
What is the average voltage across the resistor?

Q–2 In the battery-charger circuit shown below, the transformer secondary is modeled as an AC source which provides 12 V rms with an effective series resistance of $R_s = 10\ \Omega$.

($V_D = 0$ in FB)



The battery voltage V_B varies from 12 V to 14 V from the discharged to fully charged state. Consider D as an ideal diode. R_{lim} is a 50 Ω current limiting resistance placed by the designer.

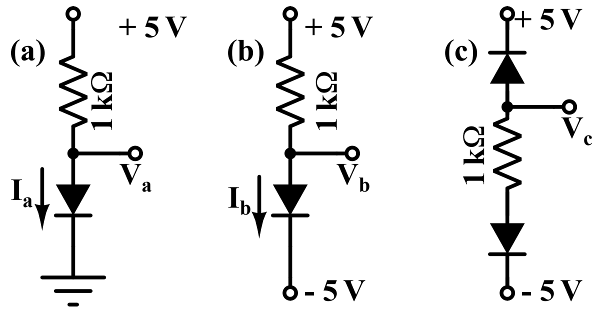
→ drop across diode = 0 in FB (ideal diode).

Sketch and label the diode current waveform when $V_B = 12$ V.
What are its peak and average values?
What are the peak and average diode currents when $V_B = 14$ V?

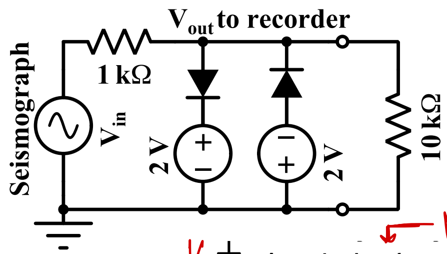
Q–3 In the three circuits (a), (b) and (c) shown below,

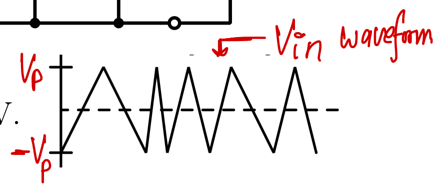
find V_a, I_a, V_b, I_b, V_c, I_c using the ideal diode model. ($V_D = 0$ in FB)

find V_a, I_a, V_b, I_b, V_c, I_c using the constant V_d model with $V_d = 0.7$ V.



Q–4 A seismograph produces an output voltage representing ground movements due to earthquakes etc. Its output is a triangular wave which goes from $-V_p$ to V_p and back, where V_p is the peak voltage. Normally its output is confined to a peak voltage of $< 1\ V$.
Its output goes to a recording instrument with an input impedance of 10 kΩ which can accept voltages up to 2 V. The recording pen will be thrown out of range and get damaged if the input voltage exceeds 3 V.
The seismograph and the recorder are stationed in a laboratory close to an armament testing facility. In rare cases during testing of explosives at the nearby facility, the output of the seismograph may shoot up to 5 V peak. We propose to use the protection circuit shown on the right.
Assume an ideal diode model. ($V_D = 0$ in FB)



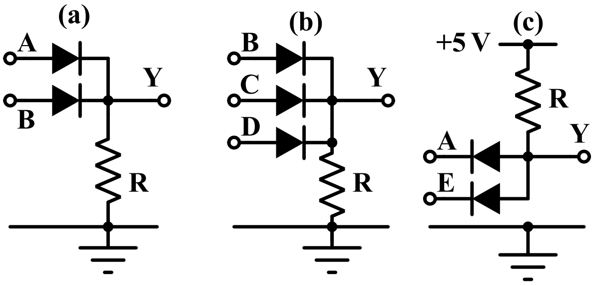


a) Plot the output waveshape in the normal case when $V_p = 1$ V.

b) Plot the output waveshape when $V_p = 5$ V.

c) How will the output look if the diode is represented by a constant V_d model, with $V_d = 0.7$ V?

Q–5 Diodes can be used to implement logic gates. In the three circuits shown below, use the ideal diode model. ($V_D = 0$ in FB)



a) What is the voltage at Y in circuits (a), (b) and (c) if $V_A = V_E = 5$ V and $V_B = V_C = V_D = 0$ V?

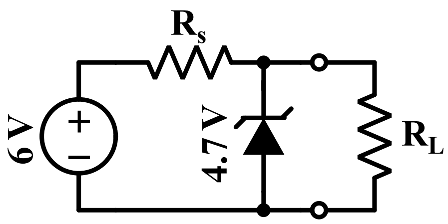
b) If 5 V represents logic '1', while 0 V represents logic '0', what is the logic function performed by the three circuits?

c) If 5 V represents logic '0', while 0 V represents logic '1', what is the logic function performed by the three circuits?

Q–6 In the reverse bias breakdown mode, the current through the diode can change over a wide range, while the voltage changes very little. This phenomenon is used to generate stable voltages.

We have a DC voltage source which supplies 6 V. We want to supply power to a circuit using this source, where the supply voltage to the circuit should be kept as close to 4.7 V as possible.

The circuit to be powered can be represented as a load resistance R_L which can vary between 1 kΩ to 10 kΩ. We use a reverse biased diode whose breakdown voltage is 4.7 V, in the circuit configuration shown on the right. Recall that in breakdown region, the current thorough the diode can vary over several orders of magnitude with very little change in the voltage across it.



a) What should be the value of the resistance R_s, such that the voltage across the load resistance is 4.7 V with the breakdown current through the diode = 1 mA when the load resistance is 1 kΩ.

b) What is the current through the diode when the load resistance is 10 kΩ.

c) What is the worst case power dissipation in R_s and the diode for values of R_L between 1 kΩ and 10 kΩ.

(When a diode is used in this way, it is called a Zener diode. In fact, that is a misnomer because the breakdown current in most diodes used for such applications is due to avalanche multiplication and not due to Zener tunneling effect. However, the name has stuck and is widely used. Indeed, the symbol used for the diode when employed for such an application resembles the letter Z as seen in the diagram above.)